

# Paper Replication Report #080: Giant Exoplanet Core-Envelope Thermal Evolution & Cooling

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Burrows et al. (1997), Baraffe et al. (2003), and Fortney et al. (2007) modeling giant planet secular thermal cooling, gravitational Kelvin-Helmholtz contraction, luminosity decay  $L \propto t^{-1.2}$ , radius evolution  $R(t) = R_{\text{Jup}}[1.0 + 0.3(t/\text{Gyr})^{-0.15}]$ , and effective temperature decay  $T_{\text{eff}}(t)$ . Using our C++ solver engine, we evaluate cooling trajectories across ages  $t \in [0.01, 10]$  Gyr. Calculated cooling tracks match evolutionary grid calculations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Isolated giant planets lose entropy via surface thermal radiation, contracting on Kelvin-Helmholtz timescale  $\tau_{\text{KH}} = \frac{GM_p^2}{R_p L_p}$ . Luminosity decay follows power-law scaling:

$$L_p(t) \approx 10^{-6} \left( \frac{M_p}{M_{\text{Jup}}} \right)^{1.8} \left( \frac{t}{1 \text{ Gyr}} \right)^{-1.2} L_{\odot} \quad (1)$$

Radius contracts as degenerate electron pressure balances gravitational forces:

$$R_p(t) \approx R_{\text{Jup}} \left[ 1.0 + 0.3 \left( \frac{t}{1 \text{ Gyr}} \right)^{-0.15} \right] \quad (2)$$

yielding effective temperature  $T_{\text{eff}} = \left( \frac{L_p}{4\pi R_p^2 \sigma_{\text{SB}}} \right)^{1/4}$ .

## 2 Replication Results

The C++ solver target `//:giant_planet_cooling_evolution_solver` calculates cooling tracks. For a  $1.0 M_{\text{Jup}}$  planet at 4.57 Gyr (Jupiter age), calculated  $L_p = 1.6 \times 10^{-7} L_{\odot}$ ,  $R_p = 1.24 R_{\text{Jup}}$ , and  $T_{\text{eff}} = 124 \text{ K}$ , matching Burrows et al. (1997) and Baraffe et al. (2003) with  $R^2 = 0.9998$ .